

FOSSILS & GEOLOGICAL TIME

Grade 4 Science Exploration Guide & Hands-on Activity

What Are Fossils?

Fossils are the **remains or traces** of plants and animals from long ago! They help scientists learn about the amazing creatures that lived millions of years before us.

- They help us learn about ancient life on Earth.
- They are found trapped in rocks and soil layers.
- They tell us the fascinating history of our planet.

What is Geological Time?

The Earth is extremely old! Geological time helps us organize and understand the long history of our planet through eras:

- **Paleozoic Era:** The age of ancient life (trilobites, early fish).
- **Mesozoic Era:** The age of dinosaurs!
- **Cenozoic Era:** The age of mammals (and humans)!

★ Why Are Fossils Important?

Fossils help us learn amazing things about Earth's history! Here is why they matter so much to scientists and explorers:

- They tell stories about how creatures lived in the past.
- They help scientists learn how life on Earth changed over millions of years.
- They show us how Earth's environments, climates, and oceans have transformed.

Did You Know? Scientists who study fossils are called *Paleontologists*!

Hands-On Activity: Let's Make a Fossil Cast!

Create your very own fossil cast using simple household materials and become a mini-paleontologist!

Materials Needed: Clay or playdough, a seashell or leaf for imprint, plaster/dough mix & water.

1. **Step 1 & 2:** Flatten your clay and press your shell or leaf firmly into it to make a deep mold, then carefully remove it.
2. **Step 3:** Mix plaster (or a simple water/dough mixture) with water until smooth.

3. **Step 4:** Pour the mix carefully into your clay imprint/mold and let it dry overnight!

Practice Questions: Fossils & Geological Time

Test your knowledge! Answer the following 12 questions below.

Student Name: _____

Date: _____

1. What is a fossil? Write a brief definition in your own words.

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2. In what two places are fossils typically found preserved?

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3. What is the scientific name for a scientist who studies fossils?

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4. Which Geological Era is known as the "Age of Dinosaurs"?

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5. Which Geological Era is known as the "Age of Mammals"?

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6. What does the Paleozoic Era represent in geological history?

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7. Give two reasons why studying fossils is important to scientists.

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8. In the fossil cast activity, what material was pressed to create the imprint mold?

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9. What are two items you can use to press into clay to make a fossil imprint?

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10. How long should you allow your fossil cast mix to dry before removing it?

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11. True or False: Fossils show us how Earth's environment has changed over time.

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12. If a scientist finds a fish fossil in a dry desert, what does that tell us about the desert's past environment?

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